

Joshua Tchou

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EDUCATION

Cornell University (GPA: 4.08)

Bachelor of Science in Mechanical Engineering, intended

Relevant coursework: Mechanics of Materials, System Dynamics, Machine Learning

Ithaca, New York

May 2028 (Expected)

SKILLS/LANGUAGES

- Python (PyTorch, MuJoCo, IsaacGym, Unitree SDK 2), Java, MATLAB
- Solidworks, Autodesk Fusion 360
- 3D printing, composite manufacturing (fiberglass, carbon fiber), machining (mill and lathe)
- Mandarin Chinese (conversational)

EXPERIENCE

Cornell Mechanical Engineering Department

Teaching Assistant for MAE 5110: Legged Robots

Ithaca, New York

Jun 2026 - Present

- Created a sim-to-real pipeline to deploy reinforcement learning locomotion policies on the Unitree Go2 robot using the Unitree Python software development kit.
- Supervised in-class activities such as assembling physical models, setting up code environments, and class discussions.

Lamplighter Lab at Cornell University

Undergraduate Researcher; Advisor: Steve Heim

Ithaca, New York

May 2025 - Present

- Wrote Python scripts to quantify the effect of control frequency on value function approximation in RL control.
- Implemented RL algorithms like REINFORCE, Monte Carlo, and TD-lambda, as well as simple 2D models such as the inverted pendulum and the gridworld using NumPy from scratch.

Cornell Math Support Center

Tutor

Ithaca, New York

Mar 2025 - Dec 2025

- Gave personalized instruction to undergraduate students in subjects including calculus, differential equations, and linear algebra.
- Worked with students in a one-on-one setting, explaining concepts and creating practice problems.

Flushing Oral & Maxillofacial Surgery

Part-time Clinical Assistant

Flushing, New York

Jun 2021 - Aug 2024

- Assisted oral surgery procedures in accordance with clinical protocols.
- Prepared for surgical procedures; sterilized surgical instruments using autoclaves, set up surgical trays.
- Maintained office cleanliness by cleaning surgical rooms, floors, and bathrooms.

EXTRACURRICULAR ACTIVITIES

Cornell Autonomous Sailboat

Subsystem Lead

Nov 2024 - Present

- Oversaw projects of ~6 team members, holding weekly meetings to align project schedules and review designs.
- Designed and manufactured mounts for the team sailboat's electronic sensors, using mill and lathe on aluminum and 3D printing.
- Designed gearbox for a camera, able to operate under wet conditions and variable orientations.
- Designed and 3D-printed model hulls to improve boat stability and reduce drag, iterating based on experimental testing and quantitative analysis.
- Used fiberglass and carbon fiber composite manufacturing techniques to manufacture hull of the boat.

Cornell Science Olympiad

Event Supervisor

Sep 2024 - May 2025

- Designed and ran middle/high school science competition events with a team of 3, supervising 100+ competitors overall.

Hall Council

Treasurer

Sep 2024 - May 2025

- Met with hall council weekly to plan and execute community events for the dorm building.
- Calculated cost of events and maintained budget.